

Akram Dweikat

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[LinkedIn](#) | [Portfolio](#)

TECHNICAL SKILLS

Cybersecurity & Networking:

NIST fundamentals, SDN security concepts, anomaly detection, packet/flow analysis, Linux & Windows OS security, network protocols (TCP/IP, DNS, HTTP, DHCP), ACLs, firewall basics, SOC fundamentals

Tools & Platforms:

Linux, Windows, macOS, Wireshark, Mininet, Ryu OpenFlow Controller, Git, Jira, Firebase, SQL Server, GCP (basic)

Programming & Scripting:

Python, SQL, Java, JavaScript, C++, C#, PowerShell (intro)

Security Concepts:

Identity & access fundamentals, threat analysis, SIEM principles, anomaly detection, cloud security basics, vulnerability awareness, secure coding practices

Soft Skills:

Client communication, problem-solving, analytical reasoning, collaboration, clear technical writing

EXPERIENCE

Software Engineer**GamBit – Halifax, Nova Scotia**

Sep 2024 – Dec 2024

- Built and optimized web and application components using JavaScript, Python, and C++.
- Collaborated with designers and developers in Agile sprints to define requirements and implement features.
- Integrated APIs, developed reusable modules, and improved performance across front-end and back-end systems.
- Participated in code reviews, unit testing, debugging, and production-quality deployments.

Data Analyst Intern

Royal Bank of Canada – Halifax, Nova Scotia

Jan 2024 – Aug 2024

- Automated and optimized data workflows using Python and SQL to streamline reporting and operations.
- Designed dashboards and validation checks to enhance usability and reliability for internal teams.
- Improved overall system performance and helped troubleshoot analytics-related issues in production environments.

Software Developer

STEMWorld Educational Services – Toronto, Ontario

Apr 2022 – Jan 2023

- Led development of interactive front-end and back-end application components using C++ and Unity.
- Implemented responsive interfaces, optimized algorithms, and improved stability of multi-user systems.
- Collaborated with QA to debug issues and improve application performance by 30%.

PROJECTS

YesChef! – AI-Powered Mobile App (React Native, Firebase, Hugging Face API)

- Developed a full-stack React Native mobile application that generates personalized recipes using Generative AI based on ingredients users already have, reducing meal-planning time for busy individuals.
- Integrated Firebase Firestore for secure authentication, user profiles, and saved recipe storage; implemented AsyncStorage for persistent sessions.
- Connected Hugging Face's AI API to produce dynamic recipe suggestions and supported native features such as camera/photo library access for uploading meal images.
- Added Google/Apple Maps integration to help users locate nearby grocery stores for missing ingredients.
- Resolved technical challenges including selecting cost-efficient AI APIs, handling image uploads, and customizing map behavior, demonstrating strong problem-solving and end-to-end mobile development skills.

SDN-Based IoT Traffic Monitoring System — Python, Ryu, OpenFlow, Mininet

- Developed an SDN system using the Ryu OpenFlow controller and Open vSwitch to monitor and classify IoT network traffic in real time, detecting anomalies and malicious behavior.
 - Extracted flow-level features (flow duration, packet/byte rates, burst patterns) from 5-tuple sequences to enable passive behavioral profiling of IoT devices.
 - Implemented unsupervised anomaly detection using DBSCAN with feature preprocessing and k-distance parameter tuning; visualized clusters to identify abnormal traffic.
 - Simulated IoT networks in Mininet and benchmarked results against BehavIoT to evaluate scalability and detection accuracy.
 - Achieved low-overhead monitoring by leveraging switch-based flow statistics, improving IoT security policy enforcement and network management.
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EDUCATION

Dalhousie University – Halifax, Nova Scotia

Bachelor of Computer Science

Certification: Communication Technologies and Cyber Security